

Compounding, Alloosemy, and Locality:

The case for root-root structures

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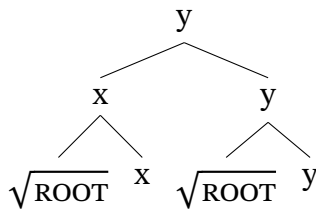
1 Introduction

In this paper I explore the issue of the locality conditions for conditioning idiosyncratic meaning in the domain of attributive compounds in English — these are compounds like *snow globe* and *tennis shoes*, in which the nonhead of the compound modifies the interpretation of the head in an unpredictable but contextually defined way. While since Arad’s influential work ([Arad 2003](#)) much investigation has been dedicated to deriving these locality constraints ([Anagnostopoulou and Samioti 2013](#); [Marantz 2013](#); [Harðarson 2016](#); [Wood 2020a,b](#)), only more recently have compounds been the focus of this type of research (see e.g. [Harðarson 2018](#); [Harðarson 2018](#); [Ingason and Sigurðsson 2020](#); [Nóbrega and Panagiotidis 2020](#)).

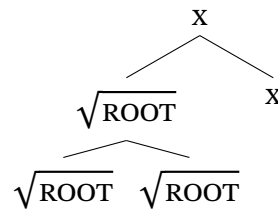
This paper contributes to both the literature on compounds and on the locality of idiosyncratic meaning by working out in depth a type of idiosyncratic meaning of subset of attributive compounds which to my knowledge has not been explored to any great extent in the literature on compounds, and which has been only briefly approached in the literature on locality ([Embick 2016](#)).

Specifically, I argue that there is a subset of attributive compounds in which the meaning of a root in the compound is conditioned by another root in the compound. I show that under the widespread assumption that roots in English compounds are categorized, most mainstream conceptions of locality do not predict post-syntactic interactions between roots in attributive compounds. In order to account for the attested interactions, I propose that there are (at least) two possible structures for attributive compounds: one which merges words (i.e. categorized roots), as in (1), and one which merges uncategorized roots directly (i.e. smaller objects than words), as in (2).

(1) **Categorized-roots structure**



(2) **Root-Root compound structure**



The paper is organized as follows. Section 2 provides some background on the locality conditions on allomorphy and allosemy. In section 3 I describe the properties of attributive compounds and argue that these compounds can be divided into subtypes on the basis of semantic differences: *mediated* attributive compounds (3.1) and *unmediated* attributive compounds (3.2). In section 4 I argue that most current theories of locality do not predict the properties of unmediated attributive compounds given our assumptions about their structure. In section 4.1 I lay out the proposal that some compounds adjoin roots, not words, and provide further motivation for this analysis. Section 4.2 deals with potential counterexamples to the proposal, while section 4.3 discusses theoretical issues concerning the status of roots as they relate to root-root structures. Section 5 concludes.

2 The locality of allomorphy and allosemy

It has long been observed that the form that syntactic objects take is dependent on context. For instance, in the context of roots like $\sqrt{\text{STAND}}$ English past tense is realized as $-\emptyset$, but in the context of other roots, such as $\sqrt{\text{LIFT}}$, it is realized $-ed$:

(3) Allomorphs of T[+PAST]

- a. $T[+PAST] \leftrightarrow /-\emptyset/ / \{\sqrt{\text{STAND}}, \sqrt{\text{CUT}}, \sqrt{\text{CATCH}}, \sqrt{\text{SING}}, \dots\} _ _$
- b. $T[+PAST] \leftrightarrow /-ed/$

Similarly, on the semantic side we find that the meaning of roots is likewise dependent on syntactic context. The root $\sqrt{\text{THROW}}$, for example, has a completely different meaning in (4a) and in (4b) — in the first case it means something along the lines of ‘toss’ or ‘fling’, whereas in the second it means ‘vomit’ (Harley 2014a:244).

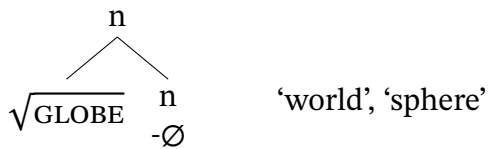
- (4) a. throw a pillow = act of yeeting a pillow
- b. throw up = vomit

The fact that the form and meaning of morphemes is dependent on their surrounding syntactic context is something that any theory of morphology needs to be able to account for. More specifically, morphological theory must be able to account for the fact that the realization of all heads is dependent on syntactic context, but not every syntactic object in a given derivation can trigger allomorphy or allosemy.

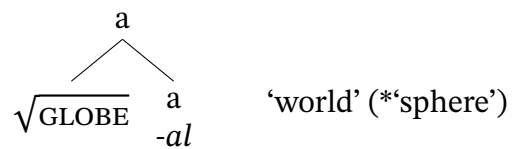
To illustrate this point, take as an example the root $\sqrt{\text{GLOBE}}$. In the context of n , as in (5), $\sqrt{\text{GLOBE}}$ can mean something along the lines of ‘world’ (as in “this troubled globe”) or ‘sphere’

(as in "Christmas globe") (Marantz 2013:104). In the context of *a* (6), on the other hand, it only has the ‘world’-related meaning; the ‘sphere’ meaning is not available. Interestingly, it is not possible for heads merged above *a* to activate the other potential meaning of $\sqrt{\text{GLOBE}}$ (i.e. ‘sphere’) — *globalize* in (7) cannot mean ‘to make sphere-like’; it *has* to build on the meaning of the adjective. Thus, the presence of *a* affects the meaning of $\sqrt{\text{GLOBE}}$ in (7), but the presence of *v* does not — *v* is not sufficiently *local* to the root to affect its meaning. It must therefore be the case that the intervening heads in each derivation are different in some important way.

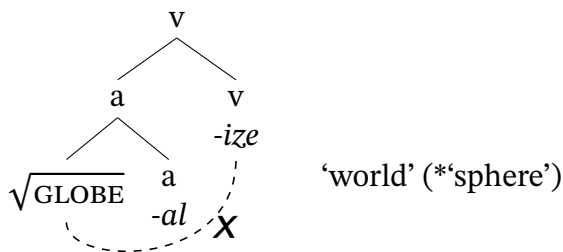
(5) *globe*



(6) *global*



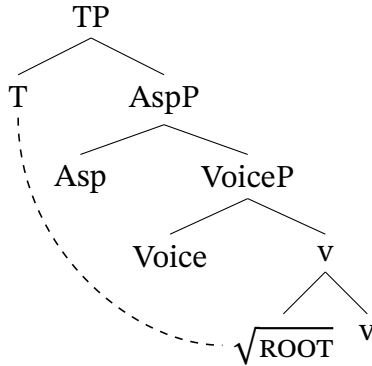
(7) *globalize*



The complexity of this issue can be further illustrated by comparing the conditioning environment of allosemy of $\sqrt{\text{GLOBE}}$ in *globalize* with the conditioning environment of allomorphy of T[+PAST], described in (3). T[+PAST] is realized as $-\emptyset$ in the context of certain roots (3a), which means that T and the root of the verb must "see" each other somehow. However, we see in (8) that in order for that to be the case T must be able to see across at least two intervening heads in the structure, Aspect and Voice. This means that T is sufficiently local to the root for its form to be affected by it, despite

the presence of more than one intervening head. This appears to contrast with *globalize*, where one intervening head was sufficient for *v* not to be local enough to the root to trigger allosemy.

(8)



In realizational theories of morphology — of which Distributed Morphology is an example — the locality conditions of allomorphy and allosemy are an issue of interpretation of syntactic structure at the interfaces (PF and LF), since it is there that syntactic objects receive phonological and semantic content. Thus, we can think of locality as involving two main components:

(9) a. *Syntactic* conditions of locality

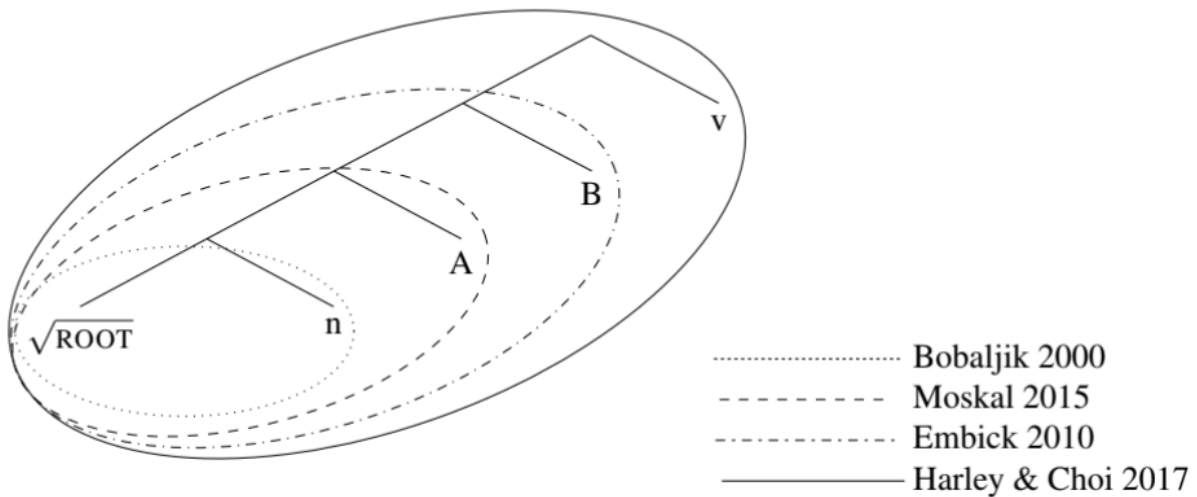
b. *Post-syntactic* or *morphological* conditions of locality

The syntactic conditions of locality determine what parts of the structure are sent for interpretation at the interfaces, while the post-syntactic conditions of locality determine what it means for a trigger of allomorphy or allosemy to "see" its target. In other words, the spellout domains established in the syntax determine the set of objects that are available for the interfaces to manipulate; which objects in the set can trigger allomorphy or allosemy in other objects is then dependent on theory-specific assumptions about operations like Vocabulary Insertion (e.g. does it occur before or after linearization?), as well as language-specific post-syntactic operations (e.g. Dissociated Node

Insertion, Impoverishment, etc).

Several locality domains have been proposed in the literature. As can be seen in the diagram in (10) these vary in restrictiveness, but with the exception of [Choi and Harley \(2019\)](#) all agree in one point: there is no conditioning of allomorphy or allosemy by outer categorizers (v in the diagram) or beyond (see also [Arad \(2003\)](#); [Marantz \(2013\)](#)).

(10) **Locality domains of contextual allomorphy** (taken from G. R. [Harðarson \(2018:90\)](#))



In this paper I will assume with [Embick \(2010, 2015\)](#) that the following conditions for allomorphy and allosemy hold:¹

1. [Embick \(2010, 2015\)](#) mainly discuss the locality conditions of allomorphy; I follow [Marantz \(2013\)](#) in taking the locality conditions of allosemy to be parallel to those of allomorphy. The linear adjacency condition can be somewhat tricky to state for the LF branch (for reasons that I will not go into detail here), but the cyclic domain restriction applies straightforwardly.

(11) For target X and trigger Y,

- a. X and Y are part of the same cyclic domain. *syntactic locality*
- b. X and Y are linearly adjacent: $X \frown Y$ *post-syntactic locality*

Cyclic domains are defined around phase heads as follows:²

(12) For cyclic head x , x 's cyclic domain includes

- a. x
- b. the complement, specifiers, and adjuncts of x
- c. edge^+ material (= noncyclic heads merged on top of the extended projection of x , up to the next phase head)

Finally,

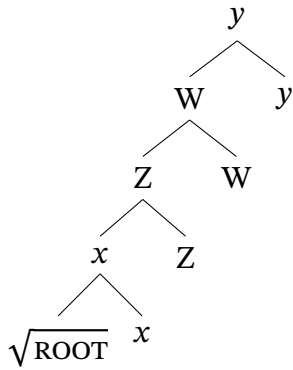
(13) Merging of cyclic heads triggers spell out of cyclic domains in their complement.

We can see how (11)-(13) work in a structure like the one in (14), where only x and y are phase heads. When x is merged with the root it triggers spell-out of all cyclic domains in its complement. Whether the root is an adjunct or a complement of x is relevant here, since per (13) cyclic heads only trigger spellout of cyclic domains in their *complement*, not in material adjoined to them. In this particular case the root is not a cyclic domain (nor does it contain any, as there is no further structure), and so even if we take it to be in the complement of x the root is not sent to spell-out at this point. Z and W are then merged to the structure without any consequences for spell-out, since they are not cyclic heads. Merging y — a cyclic head — to the structure triggers spell-out of cyclic

2. Here and throughout I will use the terms 'cyclic' and 'phase' interchangeably when describing heads and domains.

domains in its complement; if we take the structure below y to be in its complement, then the cyclic domain of x is spelled out. This includes x , the root (regardless of whether the root is a complement or an adjunct to x), and the edge+ heads Z and W . Importantly, y itself is not spelled out.

(14)



Thus, the following heads are *syntactically* sufficiently local for the purposes of allomorphy or allosemy, since they are part of the same cyclic domain:

(15) $\sqrt{\text{ROOT}}, x, Z, W$

Whether the heads in (15) are sufficiently local *post-syntactically* will depend first on how they are concatenated, and then on their content. If we assume that the structure sent for spellout is linearized into the concatenation statements in (16), then we see that insertion at the $\sqrt{\text{ROOT}}$ can only be sensitive to x as that is the only head that it is linearly adjacent to; similarly, insertion at W can be conditioned by only Z . Insertion at x on the other hand can be conditioned by the root *and* Z , and insertion at Z may be conditioned by x and W .

(16) $\sqrt{\text{ROOT}} \frown x, x \frown Z, Z \frown W$

The concatenation statements in (16) may then change depending on the concrete exponents in-

serted for each head since null nodes are removed from the structure via Pruning (Embick 2010, 2015). Thus, if a null exponent were inserted at x and Z , for instance, these heads would be deleted from the representation, leaving the remaining heads to be reconcatenated. The result of the new concatenation cycle would be linear adjacency between the root and W :

$$(17) \quad \sqrt{\text{ROOT}} \frown W$$

This is precisely what happens in cases in which the root conditions allomorphy of $T[+PAST]$ (cf. (3) and (8)). Empirically, we observe that the realization of $T[+PAST]$ is sensitive to the root of the verb *only* in cases in which v , Voice, and Asp are phonologically null. In this theory, this follows from the constraint of linear adjacency: overt heads intervene linearly between T and the root, but *null* heads are deleted from the representation, thereby satisfying the linear adjacency requirement between the two elements.

In contrast, in a word like *globalize* (cf. (7)) the realization of the root could never be sensitive to v , not because an overt head (*-al*) intervenes between the two at PF, but because a *cyclic* head (*a*) intervenes in the syntax — merging v triggers spellout of the cyclic domain of *a*, which includes the root but crucially not v .

3 Attributive compounds

Attributive compounds are compounds in which the nonhead modifies the interpretation of the head in an unpredictable but contextually determined way. Some examples of this type of compound are given in (18).

(18) **Attributive compounds**

tennis shoes, basketball, blueberry, jumper cable

In this section I argue that attributive compounds can be divided in two subgroups: one in which the head and the nonhead are interpreted via a (contextually-determined) mediating semantic relation, *R*, and one in which no such relation holds between the head and the nonhead. I refer to the former as *mediated* attributive compounds (section 3.1) and to the latter as *unmediated* attributive compounds (section 3.2).

3.1 Mediated attributive compounds

The unpredictability of the meaning of attributive compounds is nicely illustrated by the contrast in (19) between the compounds *nurse shoes* and *crocodile shoes*. As pointed out by Harley (2009), we interpret *nurse shoes* as ‘shoes *for* nurses (to wear in the course of their nurse work)’, but *crocodile shoes* we tend to interpret as ‘shoes *made of* crocodile’. But there is nothing about the structure or individual meaning of the head and the nonhead in these compounds *forcing* those interpretations — if I were to create a line of shoes especially designed for crocodiles to wear we would have no problem interpreting *crocodile shoes* as shoes *for* crocodiles (in fact, this is the most salient interpretation for a compound like *dog shoes*); likewise, it is not impossible to conceive of a (admittedly rather gruesome) world where *nurse shoes* are shoes *made of* nurse.

- (19) a. *nurse shoes* ‘shoes *for* nurses’
 b. *crocodile shoes* ‘shoes *made of* crocodile’

The actual interpretation we land on therefore depends on context and world knowledge: we

know that shoes can (unfortunately) be made of skin, but not *human* skin; we know that nurses are on their feet all throughout their shift and thus need appropriate footwear; we know that crocodiles are wild animals and that wild animals do not wear shoes, and so on.

This contrasts with synthetic compounds — compounds in which the nonhead is an internal argument of the head, as in (20). The meaning of synthetic compounds is generally predictable — a *truck driver* is someone who drives trucks, a *coffee maker* is a machine that makes coffee, and so forth. Note that no such argumental relation holds between the head and the nonhead in attributive compounds, which indicates that synthetic and attributive compounds also differ in terms of structure (Harðarson 2018; Ingason and Sigurðsson 2020; although see Harley 2009 for the opposite view).

(20) **Synthetic compounds**

a. *truck driver* ‘someone who drives trucks’

b. *coffee maker* ‘machine that makes coffee’

It is important for upcoming discussion to distinguish between the following dimensions of meaning — allosemy (21), head-nonhead relation (22), and idiomatic meaning (23). I take allosemy to parallel Vocabulary Insertion at PF — it refers to the stage in the derivation in which semantic denotations are inserted in syntactic heads.

While allosemy determines the meaning of individual heads in the structure fed to LF, what I am referring to here as the ‘head-nonhead relation’ has to do with the way those individual meanings are *composed* — just as for any other kind of structure, part of semantic interpretation is know how individual semantic denotations compose. Thus, just how the interpretation of a sentence like

Luna likes treats involves some mechanism that takes $\llbracket \text{treats} \rrbracket$ and $\llbracket \text{Luna} \rrbracket$ as being arguments of the predicate $\llbracket \text{like} \rrbracket$ (the order being given by the structure, presumably), in compounds we also need some way to know how the meaning of individual heads in a given structure compose together.

Lastly, my view on idiomatic meaning is that it involves the usual insertion of semantic denotations and consequent composition, with an added component of a convention for use. Thus, the determination of allosemantic meaning is different, and is subject to different locality conditions, than idiomatic meaning (see [Marantz 2013](#)). The focus of this paper is mainly on allosemy and the head-nonrelation; I leave idiomaticity mostly to the side.

(21) **Allosemy**

- a. Determines meaning at insertion.
- b. Example: $\sqrt{\text{GLOBE}}$ means ‘sphere’ in *snow globe*, not ‘world’.

(22) **Head-nonhead relation**

- a. Determines how the meanings of elements within the compound are composed.
- b. Example:
 - i. *nurse shoes*: shoes FOR nurses
 - ii. *alligator shoes*: shoes MADE OF alligator

(23) **Idiomatic meaning**

- a. Listed meaning, *beyond* root meaning + structural meaning.
- b. Example: *kick the bucket* involves the regular meanings of $\sqrt{\text{KICK}}$, *the*, $\sqrt{\text{BUCKET}}$, and its structure (i.e. V + object), *plus* listed ‘dying’ meaning.

In synthetic compounds like *truck driver* it seems like the meaning of the parts (i.e. *truck*, *drive*, and *-er*) plus the structure (internal argument of a root/verb) are enough to get the meaning of the whole compound right. Thus, each component of the compound has an individual meaning, and the way those meanings compose comes from the elements that are already present in the compound (these can be lexical semantics, structure, etc).

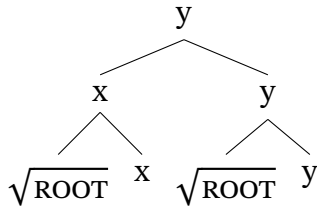
The same does not hold for attributive compounds — as we saw for *nurse shoes* versus *crocodile shoes* before, we have the meaning of the individual components of the compounds (i.e. *nurse*, *crocodile*, and *shoes*), but the precise way in which we compose these meanings is extraneous in some sense; it depends on context and world knowledge.

Different proposals have been made in the literature to account for the extra piece of meaning needed to interpret this type of compound (see [Ingason and Sigurðsson \(2020\)](#) and references therein for an overview). [Harley \(2009\)](#), for instance, argues that the difference between synthetic and attributive compounds is essentially one of lexical semantics: the two share the same type of structure, but in the case of attributive compounds there is no function-argument relation because neither the head nor the nonhead denote an argument-taking predicate.

The conflation between verb-object configurations and adjunction configurations in this proposal strikes me as problematic (see [Harðarson 2018](#)). I follow instead [Ingason and Sigurðsson](#)

(2020), who propose that attributive compounds are complex heads formed by adjunction of the head and the nonhead, as in (24).

(24) **Structure of attributive compounds (Ingason and Sigurðsson 2020)**



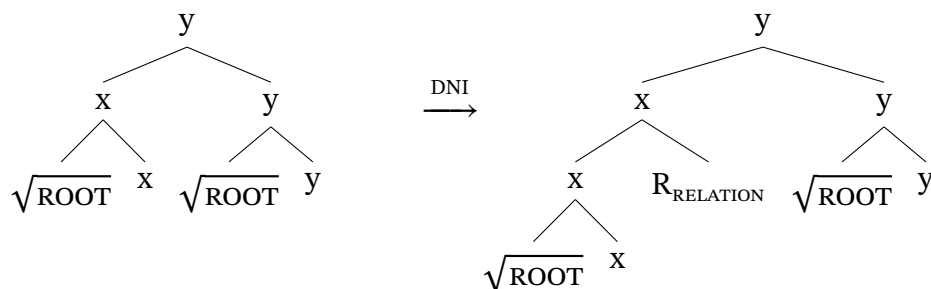
The head-nonhead relation in attributive compounds — that “extra” piece of meaning that does not seem to be readable off the structure directly at spellout — comes from the insertion at LF of a node bearing the necessary semantic relation (R), as in (25). In a compound like *nurse shoes*, for example, the R-relation is FOR; in *crocodile shoes* that relation is OF. The specific instantiation of R is selected from a restricted domain of relations (such as the nine proposed in Levi (1978): CAUSE, HAVE, MAKE, USE, BE, IN, FOR, FROM and ABOUT). The choice of relation is moreover contextually determined via Contextual Domain Restriction (CDR, Neale 1990). CDR also provides any extra information necessary for interpretation, such as such as ‘made’ and ‘skin’ in ‘shoes *made* from crocodile skin’. Thus, the structures of *nurse shoes* and *crocodile shoes* do not differ in the syntax, but the specific instantiation of the inserted R node does (and so they do differ at LF). I&S assume that both elements in primary compounds are categorized, following Embick and Marantz’s (2008) Categorization Assumption (which states that roots must merge with a categorizing head).

(25) $R_{\text{RELATION}}(X,Y) = \text{RELATION}$ holds between X and Y,

where X and Y stand for the denotations of the head and the nonhead.

The process of post-syntactic node insertion is often referred to as Dissociated Node Insertion (DNI) or node-sprouting (Choi and Harley 2019), and is generally thought to happen in cases in which certain phonological representations have no semantic effects, and vice-versa. Verbal agreement morphology in Romance, for instance, is often analyzed as involving DNI of an Agr node at PF, since the presence of agreement morphology consistently yields no effects on interpretation. The idea of inserting a dissociated R node at LF in Ingason and Sigurðsson (2020) goes along the same vein — we do not seem to find any overt realization of this relation in the morphophonology, but the presence of the R node does have important consequences for interpretation. This is schematized in (26).

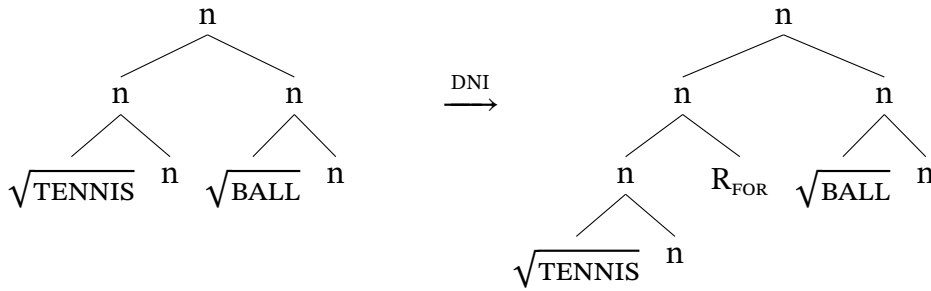
(26) **Insertion of R node at LF**



To use a concrete example, the derivation of a compound like *tennis ball* would proceed roughly as follows. Once the compound structure (on the left of the arrow in (27)) is fed to LF the insertion rule for R is triggered, and R adjoins to $[[\sqrt{\text{TENNIS}}\ n]\ n]$.³ A denotation for the specific R-relation — FOR — is given in (28).

3. There are unanswered questions here about the specifics of how R-insertion works (see "questions for future meetings" at the end of the section), but it's good enough for the purposes of this paper.

(27) **Derivation of *tennis ball* at LF**



(28) $\llbracket R_{\text{FOR}} \rrbracket = \lambda P \lambda Q \lambda x. Q(x) \wedge \text{FOR}(x, P)$

So far allosemy has not played a role in our discussion of attributive compounds; the main focus of this section has been to establish (i) that the head-nonhead relation is an important dimension of meaning for the interpretation of compounds, and (ii) that in attributive compounds this relation is dependent on context and world knowledge. In the following section I discuss cases of root allosemy driven by the presence of *other roots* in the compound.

3.2 Unmediated attributive compounds

In what follows I will argue that there is a semantic split among attributive compounds with respect to the way in which the head and the non-head compose semantically, and, relatedly, the meaning of the non-head and head roots. Specifically, I will argue that it is not the case for *all* attributive compounds that a contextually determined semantic relation holds between the head and the non-head (e.g. a tennis ball is ball *for* tennis), as proposed [Ingason and Sigurðsson \(2020\)](#). In a subset of attributive compounds no such relation exists — I refer to these as *unmediated* attributive compounds. In these compounds, such as *blueberry*, the root of the non-head does not contribute its

usual meaning (in the case of *blueberry*, the color blue). Rather, the non-head picks out a certain portion of the meaning of the head; in the case of *blueberry*, it picks out that particular type of berry (rather than any berry that is blue, for instance). .

(29) **Unmediated attributive compounds**

blueberry, pineapple, hotdog, hedgehog, breakfast, silverfish, seahorse

Consider the compounds in (30) below. The relevant differences here have to do with the non-head: in (30a) the non-head has no (synchronic) meaning; in (30b) it does have meaning, but there is no clear relation (in the sense described above) between the head and the non-head; finally, in (30c) the non-head also has meaning, but in addition there is a clear relation between the head and the non-head: a juniper berry is a berry from the juniper tree, an elderberry is a berry from the elder tree, and so on.

- (30) a. *cranberry, raspberry, mulberry*
- b. *blueberry, salmonberry, juneberry, cloudberry*
- c. *juniper berry, elderberry, yew berry, açai berry*

No such relation holds between the head and the non-head in the compounds in (30a) since the non-head has no meaning (be it a FROM relation or otherwise). The role of the non-head in these compounds is to *name* the berry — cranberries, for instance, are identifiable from other berries because of the arbitrary string of sounds that precedes *berry*. Thus, its semantic function is to point at one specific type of berry, as it were.

Despite having recognizable meaning, the function of the non-head in the compounds in (30b)

is the same as that of (30a) — to name a type of berry – and not, as might be expected, as that of (30c). Consider the data in (31) and (32) below. The relationship between the head *berry* and the non-head in the compounds in (31) is stable and relatively compositional (taking into consideration the FROM relation), and so statements such as (31a) are possible. This is not what we find for the compounds in (30b) — as shown in (32), treating the head and the non-head as being interpreted via an R relation leads to incorrect results (such as (32a)) and even to nonsensical ones in some cases (cf. (32d)).

- (31) a. all berries from the juniper tree are *juniper berries*
 b. all berries from the elder tree are *elderberries*
 c. all berries from the yew tree are *yew berries*
- (32) a. not all blue berries are *blueberries*
 b. not all June berries (i.e. berries that grow in June) are *juneberries*
 c. not all berries that bears like are *bearberries*
 d. ?? not all berries that come from clouds are *cloudberryes*

In fact, a world where red blueberries exist is easily imaginable, precisely because *blueberry* does not refer to blue berries in any significant way; it refers to a type of berry, and that berry could conceivably be a different color.

The difference between compounds like *cranberry* and *blueberry* is that the non-head roots in the latter have synchronic meaning. The particular choice of non-head seems to be some distinctive or salient characteristic or fact about the berries; thus, blueberries are so called because they are

distinctively blue, salmonberries because salmon eat them, and so on.⁴

Embick formalizes this idea in terms of the *semantic space* associated with a root (Π -space in his terminology (Embick 2016:20). A root like $\sqrt{\text{BERRY}}$ has its own Π -space, a “set of related Π meanings $\pi_1 \dots \pi_n$ ”; which parts π_n of its meaning is activated depends on the context the root occurs in. In compounds like *cranberry* or *blueberry*, the non-head points to a particular part of the Π -space of *berry*, picking out the specific type of berry in question (Embick 2016:24). Embick argues further that the effect on Π -space caused by the non-head in compounds in which the non-head has no synchronic meaning (like *cranberry*) is unidirectional, in the sense that only the semantic space of the head is affected — there is no action on the Π -space of $\sqrt{\text{CRAN}}$, since $\sqrt{\text{CRAN}}$ never occurs outside of *cranberry*. This contrasts with compounds like *blueberry*, in which the non-head $\sqrt{\text{BLUE}}$ has the same effect as $\sqrt{\text{CRAN}}$ on the Π -space of $\sqrt{\text{BERRY}}$, but $\sqrt{\text{BERRY}}$ must also have some effect on the Π -space of $\sqrt{\text{BLUE}}$, since the meaning of *blue* in *blueberry* is not compositional, as discussed above.

This sort of naming strategy is fairly widespread in English (and to my knowledge in other Germanic languages as well). Take the word *pineapple*, for instance, or *hedgehog* – both contain identifiable parts (*pine* and *apple* in *pineapple* and *hedge* and *hog* in *hedgehog*) but there is no actual relation between the two; the compounding serves only to name a new thing (in both cases by taking something that is known — *apple* and *hog*, respectively — and adding a modifier).

Thus far we have seen that mediated and unmediated attributive compounds differ semantically

4. Presumably this happens because the plant that produces these berries is not distinctive. If a berry comes from a shrub without any particularly salient features then some other characteristic is necessary to name it (if a compound name is used rather than a new word).

in at least two ways: first, in mediated attributive compounds the constituents of the compound are interpreted via an R-relation, which is absent in unmediated attributive compounds; and second, this is related to the fact that the nonhead and/or head of unmediated compounds do not contribute their “usual” compositional meaning, contrary to what happens in mediated compounds (presence and absence of “usual” meaning are represented by a check mark and an x mark in (33)).

(33) **Semantic differences**

	mediated	unmediated
R-relation	✓	✗
allosemy	✗	✓

It is worth at this point being more precise about what is meant here by “usual” and “unusual” meaning. By “usual” meaning I mean the compositional, truth-conditional meaning found for these roots outside of the context of compounds — so for instance, both $\sqrt{\text{NURSE}}$ and $\sqrt{\text{SHOE}}$ have their usual meaning in *nurse shoes*, because they mean the same thing they would generally mean in other contexts (glossing over idiomatic meanings, etc). On the other hand, the meaning of *blue* in *blueberry* is unusual since it is not the case in contexts other than this type of compound that $\sqrt{\text{BLUE}}$ has no meaning — what I’m calling the usual meaning of $\sqrt{\text{BLUE}}$ is a certain color (as in the sentence *The sky is blue*) or the feeling of sadness (as in the sentence *I’m feeling blue today*).

The crucial point here is that, if it is true that we find root meanings within these compounds that we do not find elsewhere, then it follows that the trigger for allosemy in these roots is the *other root* in the compound.⁵ And in fact that is what we find: we have already discussed the case of *blue*

5. Consider the alternatives:

i. A *categorizer* is the trigger for allosemy.

in *blueberry*, but as we can see in (34) there are plenty more examples. In *pineapple*, for instance, $\sqrt{\text{APPLE}}$ means perhaps something like ‘fruit’, which is a) not its usual meaning (that of a specific kind of fruit, as we get in *I ate an apple*), and b) not a meaning that we find for $\sqrt{\text{APPLE}}$ outside of compounds — there is no way for e.g. *I ate an apple* to mean ‘I ate a fruit’.

- (34) a. *pineapple* meaning of $\sqrt{\text{APPLE}}$?
 b. *hotdog* meaning of $\sqrt{\text{HOT}}$, $\sqrt{\text{DOG}}$?
 c. *silverfish* meaning of $\sqrt{\text{FISH}}$?⁶
 d. *deadline* meaning of $\sqrt{\text{DEAD}}$, $\sqrt{\text{LINE}}$?

Thus, we find that allosemy triggered by the other root in the compound is possible in both the head (as in the case of $\sqrt{\text{APPLE}}$, where a sort of hypernym meaning (‘fruit’) is picked over ‘apple’; see also fn.6) and the nonhead (as in $\sqrt{\text{BLUE}}$ in *blueberry*). Whatever the precise meaning of these roots is in these compounds (which can be hard to pinpoint precisely), the crucial point is that is a meaning that is *specific* to unmediated attributive compounds.

ii. The *compound structure* is the trigger for allosemy.

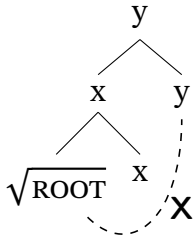
(i) implies the existence of special ‘flavors’ of *n*, *v*, *a* for compounds, which is unlikely (and undesirable). (ii) is worth considering, as compound structure may well play a role in interpretation and pronunciation (see e.g. G. R. [Harðarson 2018](#)). However, it is not clear that that role is tied strictly to *allosemy*; moreover, there isn’t to my knowledge a fleshed out theory of how that would work. I will therefore stick with root-conditioned allosemy.

6. Despite its name, a silverfish is not actually a fish but an insect. The same is true of starfish and jellyfish (where the meaning that is picked out for $\sqrt{\text{FISH}}$ seems to be more along the lines of ‘sea creature’); in contrast, a seahorse actually *is* a fish.

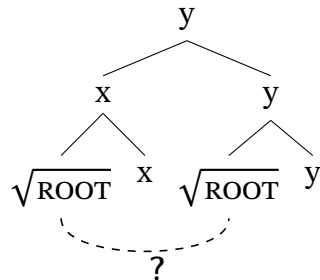
4 Enter root-root structures

In the previous section I argued that there is root allosemy in a subset of attributive compounds (unmediated attributive compounds), triggered by the presence of the other root in the compound. As we saw in section 2, both allosemy and allomorphy are subject to locality constraints — according to most current theories of locality, there is no conditioning of special meaning or form by an outer categorizer (or beyond). This is represented in (35): outer categorizer *y* cannot condition insertion at the root past *x*. The question then is whether an outer root can condition the meaning of another root in the compound, as in (36). In other words, do we predict interaction between roots in the structure proposed for attributive compounds in Ingason and Sigurðsson (2020) (cf. section 3.1)?

(35) **Outer categorizer (*y*)**

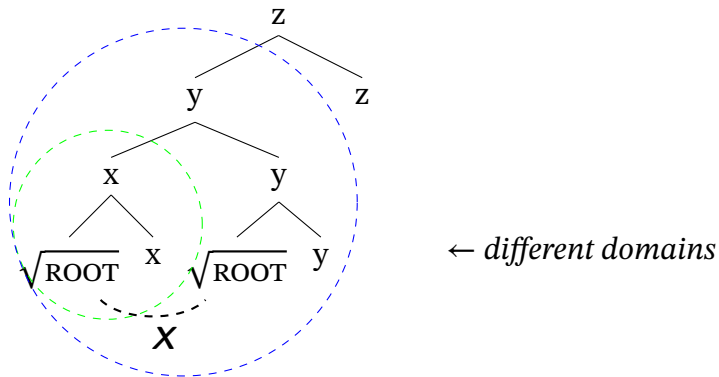


(36) **Categorized roots structure**



The answer to this question is no: the nonhead and the head roots are contained in separate cyclic domains (cf. (12)), since the nonhead categorizer (*x* in (37)) defines a cyclic domain that includes the nonhead root to the exclusion of the head. The fact that roots in unmediated attributive compounds may condition allosemy in other roots in the compound therefore remains unexplained under our current assumptions about the structure of attributive compounds (cf. section 3.1) and the locality of allosemy and allomorphy (cf. section 2).

(37) **Locality domains**



Note that this problem arises due to a combination of three distinct assumptions: (i) there is root-conditioned allosemy in some attributive compounds, (ii) allosemy is subject to locality conditions, and (iii) attributive compounds are formed by adjoining words:

(38) **Assumptions**

- a. about properties of the compounds: *root-triggered allosemy*
- b. about the locality of allosemy: *no conditioning by or after an outer categorizer*
- c. about compound structure : *attributive compounds adjoin words*

There are therefore three possible solutions to the problem, each targeting one of these individual assumptions:

(39) **Space of solutions**

- a. reject that there is allosemy in (unmediated) attributive compounds
- b. change our theory of locality
- c. change the structure of unmediated attributive compounds

I will set aside the first possibility, as I believe that the arguments presented in section 3.2 that there is allosemy in some attributive compounds are strong enough. This leaves two other options: either we (i) change our theory of locality so that it allows for the necessary interactions under the current structure, or (ii) we change the structure so that the locality conditions work out. While both approaches are reasonable, in this paper I will pursue the second — changing our assumptions about the structure of unmediated attributive compounds. I prefer it for a few reasons.

First, most theories of locality (cf. 10) do not predict an interaction between categorized roots in attributive compounds. This may of course have to do with the nature of the differences between those theories — perhaps the parameter that would need to be changed in order to account for this type of interaction is not the parameter that those theories differ along — but when multiple significantly different theories of locality agree that there should not be an interaction between roots in these compounds (under the current structure), then we should start wondering if the problem lies instead with our assumptions about structure rather than locality.

Second, changing our theory of locality is not a trivial endeavor, especially when most theories agree on precisely the point that we would be required to change. A proposal in this direction would have to allow for interactions past an outer categorizer in only a very restricted set of cases (such as the current one), but not in all the others motivating the assumptions of the previous theories. In contrast, a proposal that targets our assumptions about the structure of unmediated attributive compounds has a much more restricted scope of effect. If there are reasons to believe that such a proposal is warranted (i.e. that mediated and unmediated attributive compounds differ in structure) then on economy principles alone that is the most preferable option, at least as a first approach.

Third, the occurrence of allosemy and the absence of a mediating R-relation are not the only

ways in which mediated and unmediated attributive compounds differ; in section 4.1 I provide evidence that in addition to their special semantic properties, unmediated attributive compounds also have special phonological properties (where ‘special’ here stands for contrasting with mediated attributive compounds). The fact that the unmediated compounds cluster both special semantic and phonological properties is thus particularly suggestive of a difference in structure.

Finally, the compounds where we find this kind of special meaning generally do not have overt categorizers (although see section 4.2), which means that we do not have ‘direct’ evidence that these compounds adjoin words.

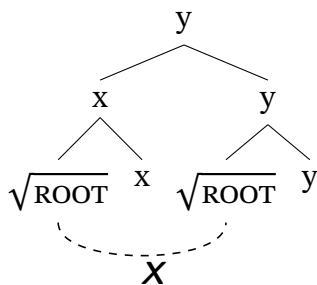
While individually these are not knock-down arguments, taken together they make a strong enough case to at least warrant a change to the structure as a first attempt. In section 4.1 I argue that there are reasons to challenge the idea that unmediated attributive compounds are formed by adjoining words, and propose instead that these compounds adjoin roots. In section 4.2 I describe what a counterexample to this proposal would look like and discuss one potential counterexample in more detail. Lastly, section 4.3 answers some arguments in the literature against root-root merger.

4.1 Unmediated attributive compounds merge roots, not words

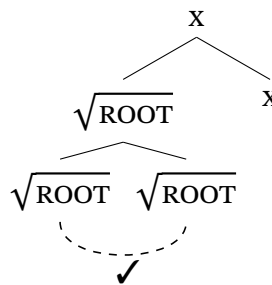
I propose that there are (at least) two different ways to form attributive compounds in English. The first, in (40), is to adjoin two categorized roots, as proposed in Ingason and Sigurðsson (2020). In this case, each root belongs to a separate cyclic domain defined by their first categorizer (cf. (37)) and thus no interaction between the head and nonhead roots is expected. This type of structure gives rise to mediated attributive compounds, which are interpreted at LF via insertion of an R node encoding a contextually determined semantic relation (cf. section 3.1). The second is to adjoin the

two roots directly, as in (41).⁷ In this case, both roots are part of the same cyclic domain since the two are categorized by the same categorizer, which means interactions between the head and nonhead roots are expected. This type of structure yields unmediated attributive compounds.

(40) **Categorized roots structure**



(41) **Root-Root structure**



This analysis does what we want in that it solves the locality problem — allosemy of $\sqrt{\text{APPLE}}$ in a compound like *pineapple*, for instance, is predicted in a structure like (41) but unexpected in a structure like (40), since in the former but not the latter the two roots are part of the same spellout domain. But the proposal would be more convincing if we had further, preferably independent evidence that a root-root structure is on the right track for unmediated attributive compounds. I turn to that evidence now.

A property that further distinguishes unmediated from mediated attributive compounds which has so far not been discussed (just alluded to at the start of section 4) is that alongside unusual semantics, some unmediated compounds also have unusual phonology (‘unusual’ in the sense described in section 3.2, i.e. not regularly found outside of compounds). Some examples of unusual phonology in unmediated attributive compounds are given in (42). The most extreme example of unusual phonology in these compounds is *breakfast*, since not only do both elements of the compound have an unusual pronunciation, but that pronunciation differs from the usual pronunciation

7. See also [Steddy \(2019\)](#) for a similar proposal.

to such an extent that it is not immediately transparent that the word contains the roots $\sqrt{\text{BREAK}}$ and $\sqrt{\text{FAST}}$ (at least not without knowledge of how to write it).

(42) **Unusual phonology** (not found outside compounds)

a.	[ˈb.ɪɛkfəst]	<i>breakfast</i>	[b.ɪɛɪkəfəst]	<i>break a fast</i>
b.	[bæ.ɪ]	<i>raspberry</i>	[bɛ.ɪ]	<i>a tasty berry</i>
c.	[lɔ̃nd]	<i>woodland</i>	[læ̃nd]	<i>a piece of land</i>
d.	[mən]	<i>chairman</i>	[mæn]	<i>a garlic man</i>
e.	[ləs]	<i>necklace</i>	[leɪs]	<i>made of lace</i>
f.	[ɪem]	<i>rainbow</i>	[ɪem]	<i>a drop of rain</i>

Other cases are less extreme, such as *necklace*, in which the head root $\sqrt{\text{LACE}}$ can be hard to recognize in certain varieties of English (like American English and British English, but not Australian English), but the nonhead $\sqrt{\text{NECK}}$ is recognizable. Cases like the reduced (i.e. schwa) pronunciation of roots like $\sqrt{\text{BERRY}}$ or $\sqrt{\text{LAND}}$ are less extreme still: both head and nonhead roots are completely recognizable.

The unusual phonology found for these compounds can be broadly described as in (43). Generally, a vowel that is usually stressed becomes unstressed, in which case it is pronounced as schwa; the exception is $\sqrt{\text{BREAK}}$ in *breakfast*, in which the vowel remains stressed but changes from tense to lax. Thus, we generally have vowel changes, most of which are accompanied by a stress shift away from the the vowel undergoing the change. This suggests that in these compounds the two roots are included in a single prosodic domain (in which the two roots are assigned stress as a unit) rather than in two separate ones. We therefore have our second piece of evidence that unmediated

attributive compounds merge roots, not words (the first piece of evidence being the allosemy facts).

(43) **Unusual phonology: generalizations**

- a. stressed $V \rightarrow$ unstressed (42a-e): $V \rightarrow \emptyset$
- b. stressed V remains stressed (42a): tense $V \rightarrow$ lax V

The next piece of evidence comes from the head-nonhead relation (or rather lack thereof) in unmediated attributive compounds. In mediated attributive compounds, the mediating R-relation is a function that takes two predicates as arguments; these predicates are supplied by the head and the nonhead (recall the example of *tennis ball* in (27) and (28): a tennis ball is a ball FOR tennis). In unmediated attributive compounds on the other hand no such relation holds (cf. section 3.2). In this analysis the absence of an R-relation in unmediated compounds actually falls out of the structure — R takes *predicates* as its arguments (type $\langle e, t \rangle$), but uncategorized roots are not predicates. Even if we allow R to be inserted post-syntactically at the root level, the roots would still not be an appropriate argument for it.⁸ Note that if one were to maintain that all attributive compounds are formed by adjoining words, then the absence of an R-relation for unmediated attributive compounds would remain unexplained; it would be necessary to appeal to something else to capture these facts.

The co-occurrence of unusual semantics, lack of an R-relation, and unusual phonology provides a further argument in favor the approach taken here (that of challenging our assumptions about the structure of attributive compounds).⁹ If there is a structural difference between mediated and

8. Minimally, there would be a type mismatch (even if we still do not have a strong handle on what the semantic type of roots is, or on what an appropriate denotation for them would look like. For some proposals in this direction, see [Pross \(2019\)](#), [Wood \(2020a\)](#), and [Song \(2021\)](#).

9. Note that unusual phonology is not attested for *all* unmediated attributive compounds; the crucial point here is that

unmediated attributive compounds, then we do expect to find both semantic and phonological differences between the two types of compound.

Moreover, the specific difference in question — direct adjunction versus intervening categorizing (i.e. phase) heads — is one that is known to lead to precisely the type of semantic and phonological differences that we find for these compounds. Take the case of *-able* adjectives, for instance (cf. [Oltra-Massuet 2014](#); [Nevins 2016](#)). High-attaching *-able* (in (44a)-(46a)) leads to predictable semantics (*X-able* = able to be Xed; inherits verb meaning) and predictable morphophonology (regular stress, regular allomorphs), while low-attaching *-able* (in (44b)-(46b)) lead to unpredictable semantics (e.g. *cómparable* means ‘similar’, not ‘able to be compared’) and unpredictable morphophonology (unusual stress, unusual allomorphs).

(44) a. compárable

b. cómparable

(45) a. eatable

b. edible

(46) a. readable

b. legible

The analysis of mediated versus unmediated attributive compounds proposed in this paper parallels that of high-attaching and low-attaching *-able* (respectively) — in mediated compounds the roots are separated by a phase boundary and so have more predictable semantic and morphophonological properties, while in unmediated compounds the roots attach directly, which means that there is no intervening phase head and thus unpredictable semantics and phonology arise.

it seems to be *only* attested for unmediated (but not for mediated) compounds.

4.2 Dealing with potential counterexamples

A counterexample to this analysis would be a structure in which the head and the nonhead are categorized and the head and/or nonhead have special meaning or phonology of the type described in sections 3.2 and 4.1, respectively. As it turns out, some compounds exist which at first glance seem to fit this description (47): many of these compounds are similar to *blueberry* in terms of meaning (e.g. *hummingbird*).

(47) **Potential counterexamples** (cf. [Steddy 2019:460](#))

- a. *anglerfish, puffer jacket, carpenter ant, killer whale, fisherman* (-er)
- b. *hummingbird, smoking jacket, weeping willow, boarding school, breaking news* (-ing)

So the question is, are *-er* and *-ing* categorizers attached to the nonhead? As an example, let's look more closely at *fisherman*. As we can see in (48)–(50) (where check-marks are for properties expected of root-root structures and X-marks for unexpected ones), the only property that is compatible with a categorized root analysis for this compound is the position of the *-er* morpheme; all the others are challenging for such an analysis (but follow from the current proposal).

(48) **Semantic properties of *fisherman***

✓allosemy in $\sqrt{\text{MAN}}$

✓non-compositionality: *fish* + *-er* + *man* ≠ 'person/man who fishes for a living'

(49) **Phonological properties of *fisherman***

✓single phonological domain: æ → ə

(50) **Morphological properties of *fisherman***

✓**fisher* does not outside of compounds¹⁰

✗*-er* appears where a categorizer *would* appear

So is a categorizer the only thing that *-er* could be? The answer to that question is no — there are at least three other ways to analyze it:

(51) **Alternative analysis for *-er* in *fisherman***

a. *-er* as a linking element.¹¹

b. More complex structure above root-root structure: *-er* is a functional head, movement accounts for surface morpheme order.

c. *-er* as part of the root: $\sqrt{\text{FISH}}$ is realized as *fisher* in the context of $\sqrt{\text{MAN}}$ (allomorphy).

The alternatives to rejecting that *-er* is a categorizer attaching to the non-head are more dramatic — one must either (i) reject the analysis, which amounts to saying that the semantic and phonological properties discussed in this paper do not point to a smaller domain (they point instead to something else explainable under current theories of locality), or (ii) reject the theory of locality, in which case unmediated attributive compounds do adjoin categorized roots (and the analysis *is* correct), but our theory of locality is wrong. All things considered (cf. section 4), admitting the

10. Note that a man who really likes fish would be a *fish man* ([mæn]), but never a **fisher man*. The same goes for a number of other possible interpretations of *fish man*, such as Aquaman-like man / a man who is somehow associated with fish (he is the source of the fish/he always talks about fish/he looks like a fish/he smells like fish/...).

11. Linking elements occur frequently in the case of *man* compounds: e.g. *hunterman*, *woodsman*, *marksman*, *kinsman*, *swordsman*, etc.

possibility of a smaller structure for compounds gets the most proverbial bang for your buck — it predicts all the types of unusual semantics and phonology that we find in unmediated attributive compounds, and it makes use of a structure that is already available to the syntax, at no cost for the theory.

4.3 Are root-root structures possible?

Since roots are objects that the syntax can manipulate in Distributed Morphology, root-root structures are part of the logical space of possibilities for structure-building; anyone looking to reject that this logical possibility is ever attested therefore requires strong arguments. In this section I present and answer some possible arguments against structures merging roots directly.

In the literature on the properties of roots there is a consensus that roots need to be categorized. This is known as the Categorization Assumption, proposed by [Embick and Marantz \(2008\)](#):

(52) **Categorization Assumption** ([Embick and Marantz 2008:6](#))

Roots cannot appear (cannot be pronounced or interpreted) without being *categorized*; they are categorized by merging syntactically with category-defining functional heads.

Building on the Categorization Assumption, [Panagiotidis \(2011\)](#) further argues that Roots can only be merged inside the complement of a categorizer, *never* of a functional head.¹²

The question then is whether this is problematic for the current proposal. That is, is the fact that roots need to be categorized (and can only be merged in the complement of a categorizer, if we

12. Panagiotidis argues that categorizers are not functional heads — they are instead the only lexical heads ([Panagiotidis 2011:366](#)). Functional heads include Num, D, T, C (etc), but not categorizers.

follow Panagiotidis) problematic for an analysis that postulates adjunction of two roots directly? I would argue that that is not the case — the Categorization Assumption would only be a problem for this proposal if roots must be immediately merged with a categorizer, but there is nothing in either [Embick and Marantz \(2008\)](#) or [Panagiotidis \(2011\)](#) to suggest that that is the case.^{13,14}

Questions arise also with respect to how root-root merger works in the syntax and the semantics. As for the syntax side, the main issues here relate to (i) our assumptions about the syntactic distribution of roots — roots have been argued to distribute as phrases (cf. [Harley 2014b:448–451](#)), as heads (cf. [Embick 2004](#); [Harley 2009, 2014a](#)), and as adjuncts (cf. [Embick 2004](#); [Marantz 2011](#)), and (ii) what to do about the fact that root-root structures involve merge of symmetrical structures, which poses a problem for labelling. Abstracting away from the first set of issues (since for all possibilities the question of root-root structures stands), I would argue that the second issue is problematic only insofar as it is problematic for adjunction in general — whatever we have to say elsewhere about how merging symmetrical structures works is what we have to say for the particular case of root-root structures in unmediated attributive compounds.

On the semantic side there is the question of how to interpret root-root structures. However, in order to know how to deal with the semantics of root-root structures we would first need to know first how to deal with the semantics of run of the mill categorized roots, which is a problem for pretty

13. Note that the Categorization Assumption states that uncategorized roots are a problem at the interfaces (PF and LF); it does not say anything about the syntax.

14. One might at this point ask whether it is possible for one categorizer categorize more than one root. I do not see any obvious reason why not (categorizers can categorize large structures, much larger than two roots; why would two roots be problematic when larger structures are not?). Moreover, nothing in Panagiotidis's analysis of categorizers seems to preclude categorization of multiple roots by a single categorizer.

much everyone.¹⁵

It is worth noting that there are also positive arguments to be made in favor of root-root structures — most prominently in this paper, root-root structures have different locality than categorized root structures, which translates into potential differences at the interfaces. As I argue in section 4.1, appealing to root-root structures for unmediated attributive compounds capitalizes precisely on the different locality conditions for roots in those compounds as opposed to roots in mediated compounds (which are categorized prior to adjunction). In fact, root-root structures have also been argued to account for properties of certain compounds in other languages, such Dutch (De Belder 2017), English (Steddy 2019), Ojibwe (Newell and Piggott 2014), Yixing Chinese (Hu and Perry 2018), and Mandarin Chinese (Zhang 2007, Lai 2016, Cheng and Liu 2020).¹⁶

Moreover, there is another type of compound in English for which it would be hard to argue against a root-root analysis — the so-called neoclassical compounds (Bauer 1998). These are compounds formed out of bound Greek and Latin roots, such as *astronaut*:

(53) **Neoclassical compounds**

astronaut, democracy, thermometer, biology, autophagy, allophony, phonology, oligarchy,...

These really look like root-root compounding: $[\sqrt{\text{ROOT}} + \sqrt{\text{ROOT}}] + o$ (inserted between the

15. See Pross (2019), Wood (2020a), and Song (2021) for some proposals tackling this issue.

16. Although see Song (2017) for an opposing view on Mandarin Chinese compounds.

two roots).¹⁷ First, neoclassical compounds mostly contain bound roots.¹⁸ Second, the types of root allomorphs that we find for these compounds resemble those of low-attaching *-able* discussed in section 4.1 (cf. (45b) and (46b)). Lastly, the meaning of some of these roots is quite hard to define (for instance, what exactly do $\sqrt{\text{CRAC}}$ in *democracy* and $\sqrt{\text{ARCH}}$ in *oligarchy* mean? Something along the lines of ‘related to power’? Are they synonyms?), in a way not entirely dissimilar from roots like $\sqrt{\text{CRAN}}$ and $\sqrt{\text{RASP}}$ in compounds headed by $\sqrt{\text{BERRY}}$.

Thus, it does not appear as though there are any knock-down arguments against root-root structures. On the other hand, the advantages of including them in our structure building are significant.

5 Conclusion

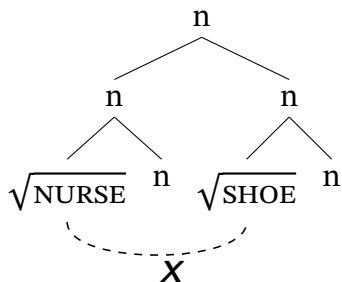
In this paper I have argued that there are two distinct types of attributive compound: mediated and unmediated. The two differ along semantic and phonological properties: in mediated attributive compounds there is a mediating semantic relation (R) between the head and the non-head, no root allosemy, and two distinct prosodic domains; in contrast, unmediated attributive compounds lack a mediating R-relation, do have root allosemy, and have only one prosodic domain for stress assignment (cf. sections 3.2 and 4.1). The properties of the latter type of compound are not predicted under common assumptions about the locality of allosemy (cf. section 2) and attributive compound structure (cf. section 3.1). I propose instead that the differences between mediated and unmedi-

17. One could argue that *o* is a categorizer, but it could likewise be argued that it is part of the root itself, or (perhaps more likely) that the vowel is inserted in the context of Greek roots in C-C environments.

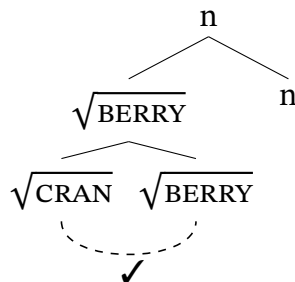
18. Some may have become words in their own right in English, such as *rhyno* (clipping from *rhinoceros*) and *meter* (potentially because it is also unbound in French, where it was borrowed from).

ated compounds are due to differences in structure — mediated compounds like *nurse shoes* merge words, while unmediated compounds like *cranberry* merge roots:

(54) **Structure of *nurse shoes***



(55) **Structure of *cranberry***



Since we take roots to be individual elements in the syntax that exist independently of categorizers, merging two roots directly is one of the possible options the grammar has access to. I have argued that the arguments that have been made against this possibility do not in fact preclude it (cf. section 4.3). This analysis of the compounding facts therefore adds another empirical argument to the usefulness of root-root structures.

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